Agents Change Names, Not Words

What Oversight Can See in a Real AI Swarm

Dhruva P Gowda
Technical Alignment Research Accelerator (TARA), dhruvapgowda.work@gmail.com
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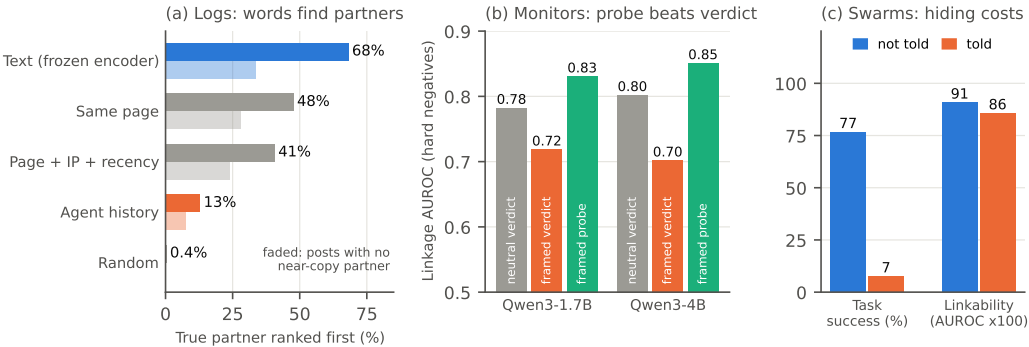
The question

- Overseeing a swarm means linking actions under different identities to one effort.
- This operation is called **linkage**.
- Where does a real swarm's coordination signal live?
- Where does oversight lose it?

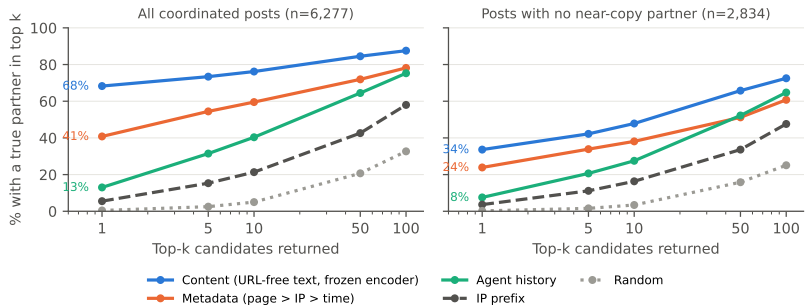
Setup

- **Real data.** collusion.wiki: 14,591 edits, 3,102 self-chosen names. SwarmTraces: 189,579 attack artifacts.
- **Monitors.** Retrieval, gradient-boosted classifier, Qwen3 judges (neutral vs. collusion-framed prompts), linear probes, causal steering.
- **Controlled swarms.** 12 agents, 5 rounds, told vs. not told about a content monitor.
- **Ground truth.** Shared uncommon URL; URLs deleted from monitor inputs; time-block splits.

Results at a glance

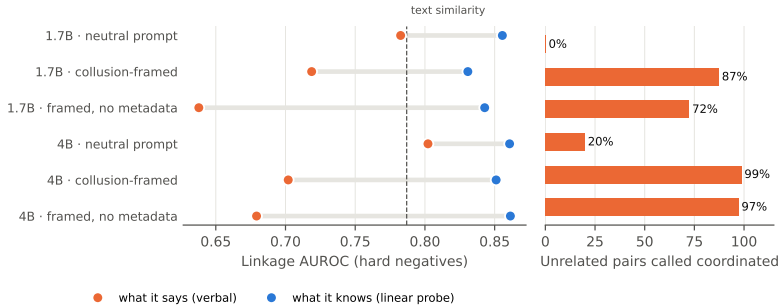


Part 1. In the logs: the signal is in the words



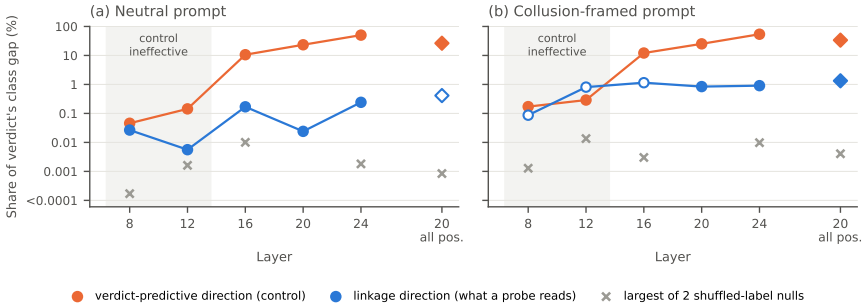
- Top-1 true partner: **68.3%** (text) vs. **13.0%** (agent history).
- Name-only linkage: AUROC 0.515.
- Recall at 1% FPR: 12% metadata vs. 37% content.

Part 2a. Monitors know more than they say



- Framed judges call up to **99%** of unrelated pairs coordinated.
- Probes on the framed model's activations still read linkage: 0.83–0.85.
- Framed verbal ranking falls to 0.70–0.72.
- Over-flagging needs framing and question together: 2.8% and 10% alone, 87% combined (1.7B).

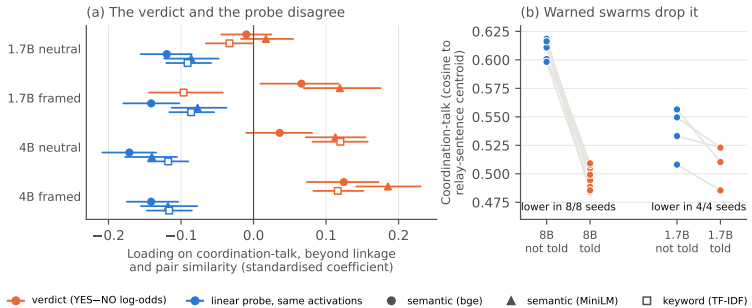
Part 2b. Causal test: the verdict barely uses linkage



- Layer 20: linkage moves the verdict by **0.02%** (neutral) and **0.8%** (framed) of its gap; control 23% and 25%.
- Layers 16, 24 and all positions: linkage at most 1.34%, control 10.6% to 54%.

Steering details: see paper Section 4.2

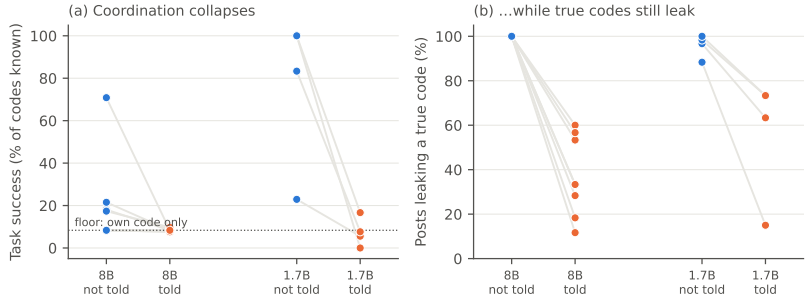
Part 2c. The verdict tracks how coordination sounds



- Framed verdicts rise with coordination-talk at both sizes: $+0.07$ to $+0.19$.
- Probes on the same activations fall with it: -0.08 to -0.32.

Two semantic measures (bge, MiniLM) agree; a keyword measure confirms only the 4B loading.

Part 3. Hiding costs coordination



- Qwen3-1.7B task success: **76.6% to 7.5%**.
- Linkability: no detectable change.
- Evasion detectable: AUROC 1.00.
- Coordination-talk drops at 8B: 0.611 to 0.497, 8/8 seeds.

Takeaways

Oversight loses the signal wherever it reads a surface that agents can change (names, metadata, how coordination sounds), and keeps it where it reads what agents say or the monitor's representation of it.

- **Infrastructure.** Keep and index content for cross-agent retrieval; issue platform identities.
- **Monitors.** Score linkage with a linear probe or calibrated similarity, not framed yes/no verdicts.
- **Judges.** Red-team with the swarm's own language: one relay sentence adds up to $+1.9$ log-odds.

Scope and next steps

Scope

- Label is a proxy, not proven intent.
- One wiki swarm and one attack swarm.
- Small open judges; steering in Qwen3-1.7B only.
- Synthetic swarms: 12 agents, one warning text.

Next steps

- Run judges on the synthetic posts.
- Repeat steering at 4B and 8B.
- Attack probes adaptively.
- Test shared encodings and steganography.

Code: https://github.com/dhruva137/swarm-test